

ChE 417 – Data, Statistics, Modeling for Engineers and Scientists

Homework 1 – (10 points)

Problem 1 – plot CDF and PDF for exponential RV

4 points

The exponential RV model is widely used in survival analysis (e.g. RV represents the time to wait until an asset fails). This model has the pdf:

$$f(x) = \frac{1}{\beta} * \exp\left(-\frac{x}{\beta}\right), D = [0, \infty), \beta = \text{parameter}$$

Questions –

1. For $\beta = 0.5, 1$, and 1.5 - Plot PDF and CDF. Determine the CDF from the PDF mathematically.
2. Verify if the PDF and CDF satisfy the necessary conditions for the RV model (check continuous or discrete first).

Problem 2 – uncertainty propagation and time value of money

6 points

Present value PV of cash flow C (revenue) obtained at a future time t is given by:

$$PV = \frac{C}{(1+i)^t}$$

Here, i = interest rate. Assume $i = 0.1$ (10%). The term $\alpha = \frac{1}{(1+i)^t}$ is known as the discount factor, which relates the present and future value of money. Note that the discount factor is less than or equal to 1, and thus present value is always less than or equal to future value. In other words, value of money increases over time.

$$C = \frac{PV}{\alpha}$$

Cash flow C that you expect from an investment is uncertain, and a financial analyst has obtained a number of possible scenarios for you – (refer to cash_flow.dat)

1. Visualize the PDF of C and use Monte Carlo simulations to obtain the PDF for the present value PV for times $t = 1, 5$, and 10 years. Use MC to obtain the mean and standard deviation of PV as a function of time. Discuss how these summarizing statistics and the PDF vary over time. (Python function for MC - `np.random.normal()`, use a set seed value - `np.random.seed(0)`)
2. Assume C is a continuous RV and derive a formula for the uncertain SD of PV as a function of alpha and of the SD of cash flow. Here, assume $Y = \phi(x)$, calculate $\mathbb{E}(X)$, $\mathbb{E}(Y)$, $\mathbb{V}(X)$, $\mathbb{V}(Y)$, where, $Y = PV$ and $X = C$. Relate $\mathbb{V}(X)$ and $\mathbb{V}(Y)$ and correspondingly $\mathbb{SD}(X)$ and $\mathbb{SD}(Y)$.
3. Find what happens to uncertainty of PV ($\mathbb{SD}(Y)$), when the time is long i.e., $t \rightarrow \infty$
4. What happens to uncertainty in PV ($\mathbb{SD}(Y)$), when interest is small, i.e., $i \rightarrow 0$